

Relational Clock Transport in Connected Jacobi Systems: Complete Observables and Clock-Fiber Geometry

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Abstract

We study transport of compact phase clocks in a connected, reparameterization-invariant Jacobi system. For N clocks with positive fiber inertias $I_i(u)$, conserved phase charges and conditioning on monotonic organization surfaces define branch-local complete observables. The endpoint transport factors obey

$$\ln \Xi_{ij} = d_j - d_i, \quad d_i = \ln I_i(u_+) - \ln I_i(u_-).$$

Consequently, only $N - 1$ transport factors are independent, and the observable response is the class of $d \in \mathbb{R}^N$ in $\mathbb{R}^N / \text{span}\{(1, \dots, 1)\}$. A common fractional response is therefore absent from every local pairwise comparison. We prove that a nonconstant clock-fiber circumference cannot be removed by a degree-one, orientation-preserving diffeomorphism that preserves the specified 2π clock cycle. A positive graph-Laplacian block separately measures changes of active particle-separation records, and its coefficient is shown to increase the stationary Jacobi action on each regular fixed-endpoint branch. Finally, a stable fast-mode model with constant bare kinetic coefficients is constructed. Adiabatic restriction to its equilibrium manifold induces both organization-dependent clock inertias and a corrected organization metric, showing that the clock-fiber functions need not be inserted directly into the bare kinetic geometry. The results concern finite constrained mechanics and do not assume a spacetime or gravitational interpretation.

Keywords: Jacobi mechanics; relational clocks; complete observables; constrained systems; configuration-space geometry; graph Laplacians.

1 Introduction

A physical clock is a material subsystem whose reading is inferred from a repeatable record. In a generally covariant or reparameterization-invariant description, such a record can label correlations without supplying an external physical time. Jacobi actions, best matching, partial and complete observables, and changes of internal clock provide established frameworks for making those correlations gauge invariant [1–7].

A specific calibration problem arises when the organization of an interacting system changes. Suppose a clock is assigned one effective kinetic coefficient in an open interaction network and another in a closed network. If the two organizations are represented by disconnected models, each model admits an independent calibration. A comparison then has no invariant content until the clock cycles and canonical data have been identified across the two models. Equal momenta, equal rates, and equal clock energies are distinct preparation rules and generally produce different endpoint ratios.

The mathematical question is therefore

Under what conditions does an interaction-dependent clock comparison survive gauge freedom, independent calibration, and preparation ambiguity as a relational observable?

Conservation of phase momentum with a variable inertia already gives the familiar relation $\dot{\theta} \propto I^{-1}$. The issue considered here is instead whether a comparison based on that relation refers to the same compact clock orbits at two relationally selected events, uses one specified transport protocol, and survives both gauge reduction and admissible clock-preserving coordinate changes.

Three distinctions are essential. First, a gauge transformation changes the description of one physical history, whereas resetting a clock changes the history or its preparation. Second, a coordinate redefinition of a compact phase is not automatically an equivalence of clock constructions: one physical cycle is part of the operational identification. Third, persistent interaction organization and departure from particle co-motion are different structures. A network may reorganize while particle separations remain fixed, and particles may change their relational separations within a fixed organizational state.

We embed the organizations in one connected configuration space and transport the same compact clocks through a continuous interaction-link activation. Their phase momenta are Noether charges of one action. Departure from particle co-motion is represented separately by a positive graph-Laplacian quadratic form on relational differentials. Thus an arbitrary cross-sector calibration rule is replaced by transport on one gauge orbit. Uninterrupted transport, endpoint rate resetting, and equal-energy preparation nevertheless remain different physical protocols. Dynamical interaction links place the model near the broad class of adaptive networks, although the present construction is finite mechanical rather than a network-evolution model [8].

Theorem 5 gives the transport result for N clocks. Proposition 6 identifies the $N - 1$ independent observables with a quotient of the logarithmic response space by its common-response direction. Proposition 7 proves that variable fiber circumference is not removable by a degree-one clock-preserving diffeomorphism. A variational result shows that the stationary abbreviated action is nondecreasing in the coefficient of the positive active-link block on every fixed regular branch. Proposition 2 then supplies an explicit stable fast-mode model whose adiabatic equilibrium manifold induces variable clock inertias from a constant bare kinetic metric.

The scope is finite. The induced-metric construction is a solvable mechanical completion, not a microscopic model of an atomic or optical clock. No universal temporal standard, field-theory constraint algebra, quantum dynamics, or metric theory of gravity is inferred. The purpose is to characterize the gauge-invariant transport data of connected clock fibers and to distinguish that data from the positive geometry of particle non-co-motion.

2 Relational clocks, gauge, and preparation

2.1 Inherited machinery and the restricted claim

A degree-one Jacobi action is already a geodesic principle on configuration space [9]. Best matching already removes designated configurational redundancies, and complete observables already construct gauge-invariant correlations in constrained systems [3, 4]. Clock-neutral formulations explain how distinct reduced clock descriptions can be related through a common constrained description [6, 7]. Relational particle models provide established finite arenas for these issues [10]. Weighted graph Laplacians and their consensus null spaces are likewise standard structures in spectral graph theory and network dynamics [11, 12].

The proposed contribution is not a new graph-theoretic null-space theorem, a new general theory of complete observables, or the identity $I\dot{\theta} = \text{const}$. It is the combined result obtained by placing two interaction-dependent blocks in one connected relational Jacobi metric: a clock-fiber block that responds to persistent organization and an active-link block that responds to changes of particle-separation records. The normalized clock cycles, conserved canonical data, and preparation protocol are carried through one constrained history. The resulting N -clock transport data have a quotient structure not visible in a single variable-inertia equation. Table 1 separates inherited tools from the additions.

Element	Status in this paper
Degree-one Jacobi action and Hamiltonian constraint	Established relational-mechanics structure
Translation best matching and complete observables	Established constrained-system framework
Weighted graph Laplacian and connected-component null space	Established graph-theoretic structure
Variable-inertia identity $I\dot{\theta} = \text{const}$	Elementary mechanical input, not the novelty
Continuous interaction-link coordinate in one phase space	Model choice introduced here
Interaction-dependent compact clock fibers	Proposed persistent-organization coupling
Active-link quadratic form on relational differentials	Proposed particle-non-co-motion coupling
Same-orbit transport with fixed clock identity and protocol	Principal relational-clock construction
Gauge-invariant endpoint theorem and clock-preserving non-removability	Principal mathematical results
N -clock calibration quotient and induced fast-mode metric	Principal mathematical results
Universal clocks, spacetime locality, and metric gravity	Outside the finite construction

Table 1: Established structures and the restricted contribution of the model.

Because the clocks used below are compact phases, each is a locally monotonic record only after one cycle has been lifted to the universal cover. Periodic clocks require greater care than globally monotonic clocks, and naive winding counting does not supply a global Dirac observable without additional structure [13]. All complete-observable statements below are therefore branch local.

2.2 Operational clock identity

A clock construction consists here of a compact orbit, its orientation, and the identification of one complete orbit with one operational cycle. For the i th clock this is a specified $U(1)$ action generated by

$$k_i = \frac{\partial}{\partial \theta_i}, \quad \theta_i \sim \theta_i + 2\pi. \quad (1)$$

A shift of phase origin, an orientation-preserving reparameterization of the system orbit, or a global particle translation does not change that clock. A transformation that changes which closed curve counts as one cycle changes the clock construction rather than merely its coordinates. This distinction is used in Sec. 7 to separate clock-preserving equivalences from physical recalibration.

2.3 Preparation is physical input

A clock comparison is incomplete until its preparation protocol is specified. In the connected model, the natural protocol is uninterrupted transport: the same two clock degrees of freedom evolve on one gauge orbit with no external intervention on their $U(1)$ charges. An independently prepared endpoint experiment can instead impose equal angular rates or equal individual clock energies. These preparations are not gauge choices. They alter canonical data and therefore answer different physical questions.

3 General connected Jacobi construction

Let q^A be dimensionless coordinates on a configuration manifold Q . A group G represents descriptive redundancy, with infinitesimal generators $\xi_\alpha^A(q)$. The best-matched differential is

$$Dq^A = dq^A - dg^\alpha \xi_\alpha^A(q). \quad (2)$$

Let $G_{AB}(q)$ be a positive metric and

$$\Omega(q) = E - V(q) > 0 \quad (3)$$

throughout the history. The Jacobi action is

$$S_J = S_0 \int d\lambda \sqrt{2\Omega(q)G_{AB}(q)D_\lambda q^A D_\lambda q^B}. \quad (4)$$

Homogeneity of degree one makes the arbitrary label λ physically irrelevant. An equivalent einbein action is

$$S_E = \frac{S_0}{2} \int d\lambda [N^{-1}G_{AB}D_\lambda q^A D_\lambda q^B + 2N(E - V)]. \quad (5)$$

With dimensionless momenta $p_A = P_A/S_0$, variation with respect to N gives

$$\mathcal{H} = \frac{1}{2}G^{AB}p_A p_B + V - E \approx 0, \quad (6)$$

while free-endpoint variation of the best-matching variables gives

$$\mathcal{J}_\alpha = p_A \xi_\alpha^A \approx 0. \quad (7)$$

3.1 Persistent organization and particle co-motion

Let u be invariant under the configurational gauge group G and describe a continuous interaction change. It is not, in general, a Dirac observable of the Hamiltonian constraint: under Hamiltonian flow it is a partial observable that can serve as a branch-local relational label. Let $C(u)$ be a coarse-grained organization descriptor entering an effective metric. No identification with entropy, algorithmic complexity, or a complete microscopic interaction count is assumed.

Let $R_e(q)$ denote the particle-separation record carried by an active interaction channel e . Exact particle co-motion on a history segment means

$$DR_e = 0 \quad (8)$$

for every active channel on that segment. This definition invokes neither coordinate velocity nor a preferred frame. For nonnegative active-link weights $w_e(q)$, define

$$dM^2 = \sum_e w_e(q)(DR_e)^2. \quad (9)$$

This form is positive semidefinite and vanishes exactly when all positively weighted particle-separation records are unchanged. For a graph with incidence matrix B and diagonal weight matrix W ,

$$dM^2 = dy^T L dy, \quad L = B^T W B. \quad (10)$$

Proposition 1 (Active-link co-motion null space). *For nonnegative link weights, $dM^2 = 0$ if and only if the particle displacement is constant on each connected component of the positively weighted graph. If the active graph is connected, the only null direction is global translation, and dM^2 is positive definite after translation reduction.*

Proof. The form is a sum of nonnegative weighted squares. It vanishes exactly when the displacement difference on every positively weighted edge vanishes. Equality propagates along every path in a connected component. The converse is immediate. $\square$

Persistent organization and particle co-motion are therefore logically distinct. The link variable u and organization scalar $C(u)$ may change while $DR_e = 0$. Full network stasis is the stronger condition $DR_e = 0$ for all active edges together with $du = 0$.

3.2 General compact clock fibers

Take $N \geq 2$ compact phase clocks with arbitrary positive inertias,

$$\theta_i \in S^1, \quad I_i(u) > 0, \quad i = 1, \dots, N. \quad (11)$$

Their clock-sector line element is

$$d\Sigma_{\text{clock}}^2 = \sum_{i=1}^N I_i(u) d\theta_i^2. \quad (12)$$

The transport theorem is derived for these general functions before any coupling ansatz is imposed. The explicit network model below uses $N = 2$ so that its preparation protocols and numerical orbit can be displayed without additional indices.

4 A minimal interaction-network model

4.1 Configuration space and continuous link activation

Consider three distinguishable particles on a line with coordinates y_a , $a = 1, 2, 3$. The links 12 and 23 are permanently active. A dimensionless coordinate $u \in \mathbb{R}$ controls the strength of the additional 31 channel. The organization window is

$$0 \leq u \leq 1, \quad u = 0 \text{ open chain}, \quad u = 1 \text{ fully activated closure link}. \quad (13)$$

The process is a continuous activation of one interaction channel, not a singular change in the topology of configuration space. The kinematic space is

$$Q = \mathbb{R}_y^3 \times \mathbb{R}_u \times S_{\theta_1}^1 \times S_{\theta_2}^1. \quad (14)$$

Only global translations are gauge:

$$y_a \mapsto y_a + \epsilon, \quad u \mapsto u, \quad \theta_i \mapsto \theta_i. \quad (15)$$

No claim of scale, reflection, particle-relabeling, or full Euclidean-frame invariance is made.

On the physical branch, assign

$$w_{12} = 1, \quad w_{23} = 1, \quad w_{31} = u^2. \quad (16)$$

The active-link form is then

$$dM^2 = dy^T L(u) dy = (dy_1 - dy_2)^2 + (dy_2 - dy_3)^2 + u^2(dy_3 - dy_1)^2. \quad (17)$$

The permanent links keep the graph connected throughout $0 \leq u \leq 1$. For an explicit organization descriptor, choose

$$C(u) = 2 + (1 + \kappa)u^2, \quad \kappa > -1. \quad (18)$$

The constant term records the two permanent channels, and the nonnegative u^2 contribution records the activated closure channel and its cycle contribution. This simple expression is a model choice, not a uniqueness claim.

4.2 Effective phase-clock ansatz

To embed the phase metric in a larger effective clock sector, introduce complex variables

$$\psi_i = r_i e^{i\theta_i} \quad (19)$$

with clock-sector kinetic line element and radial potential

$$d\Sigma_{\text{clock}}^2 = \sum_{i=1}^2 [M_i + g_i C(u)] d\psi_i^* d\psi_i, \quad (20)$$

$$U_{r,i}(r_i) = \frac{m_{r,i}^2}{2} (r_i - r_{i0})^2 + O[(r_i - r_{i0})^3]. \quad (21)$$

When the radial modes are heavy relative to the organization-change scale, restriction to the low-energy phase submanifold gives

$$I_i(u) = r_{i0}^2 [M_i + g_i C(u)], \quad M_i + g_i C(u) > 0. \quad (22)$$

The fractional response is

$$\chi_i(C) = \frac{d \ln I_i}{dC} = \frac{g_i}{M_i + g_i C}. \quad (23)$$

This is an effective kinetic-coupling ansatz. Whether realistic clocks possess such couplings, or flow toward a universal low-energy response, is not decided here.

4.3 Induced clock geometry from constant bare masses

The functions $I_i(u)$ need not be placed directly in the bare kinetic metric. The following finite model gives a solvable induced-metric construction. Introduce radial clock coordinates $r_i > 0$ and a mediator $\zeta \in \mathbb{R}$, with constant-mass kinetic line element

$$d\Sigma_{\text{bare}}^2 = K du^2 + m_\zeta d\zeta^2 + \sum_{i=1}^N m_i (dr_i^2 + r_i^2 d\theta_i^2), \quad K, m_\zeta, m_i > 0. \quad (24)$$

Let $\mathcal{C}(u)$ be a smooth organization function and set

$$\begin{aligned} U_f(r, \zeta; u) = & \frac{1}{2} \sum_{i=1}^N k_i (r_i - r_{i0})^2 + \frac{\kappa_\zeta}{2} [\zeta - \mathcal{C}(u)]^2 \\ & - \zeta \sum_{i=1}^N \lambda_i (r_i - r_{i0}) + \frac{\gamma}{2} \mathcal{C}(u)^2, \end{aligned} \quad (25)$$

where $k_i, \kappa_\zeta > 0$. Define

$$\boldsymbol{\lambda} = (\lambda_1, \dots, \lambda_N)^T, \quad \Lambda = \sum_{i=1}^N \frac{\lambda_i^2}{k_i}, \quad \Delta = \kappa_\zeta - \Lambda. \quad (26)$$

Proposition 2 (Adiabatically induced clock metric). *Suppose $\Delta > 0$, $\gamma > \kappa_\zeta \Lambda / \Delta$, and $r_{i*}(u) > 0$ on the interval of interest. The fast potential (25) has a unique stable equilibrium at*

$$\zeta_*(u) = \frac{\kappa_\zeta}{\Delta} \mathcal{C}(u), \quad r_{i*}(u) = r_{i0} + \frac{\lambda_i \kappa_\zeta}{k_i \Delta} \mathcal{C}(u). \quad (27)$$

Restriction of the bare metric (24) to this equilibrium manifold gives

$$d\Sigma_{\text{ind}}^2 = K_{\text{eff}}(u) du^2 + \sum_{i=1}^N I_i^{\text{ind}}(u) d\theta_i^2, \quad (28)$$

where

$$I_i^{\text{ind}}(u) = m_i r_{i*}(u)^2, \quad (29)$$

$$K_{\text{eff}}(u) = K + [\mathcal{C}'(u)]^2 \left[m_\zeta \left(\frac{\kappa_\zeta}{\Delta} \right)^2 + \sum_{i=1}^N m_i \left(\frac{\lambda_i \kappa_\zeta}{k_i \Delta} \right)^2 \right]. \quad (30)$$

The effective equilibrium potential is

$$U_{f,*}(u) = \frac{1}{2} \left(\gamma - \frac{\kappa_\zeta \Lambda}{\Delta} \right) \mathcal{C}(u)^2. \quad (31)$$

Proof. The Hessian of U_f in the fast variables (r_i, ζ) is the block matrix

$$H_f = \begin{pmatrix} \text{diag}(k_i) & -\boldsymbol{\lambda} \\ -\boldsymbol{\lambda}^T & \kappa_\zeta \end{pmatrix}. \quad (32)$$

Its Schur complement with respect to $\text{diag}(k_i)$ is Δ . Hence H_f is positive definite precisely when $\Delta > 0$. Solving $\partial_{r_i} U_f = 0$ and $\partial_\zeta U_f = 0$ gives (27). Substitution gives (31). Pulling (24) back along the equilibrium embedding gives (28)–(30). Positivity follows from the stated assumptions. $\square$

Thus organization-dependent clock-fiber circumferences arise here from the embedding of a stable equilibrium manifold in a flat constant-mass clock sector. Writing

$$\beta_i = \frac{\lambda_i \kappa_\zeta}{k_i \Delta r_{i0}}, \quad I_i^{\text{ind}}(u) = m_i r_{i0}^2 [1 + \beta_i \mathcal{C}(u)]^2, \quad (33)$$

the induced pairwise endpoint factor of Theorem 5 is

$$\Xi_{ij}^{+-} = \left[\frac{[1 + \beta_j \mathcal{C}(u_+)] [1 + \beta_i \mathcal{C}(u_-)]}{[1 + \beta_i \mathcal{C}(u_+)] [1 + \beta_j \mathcal{C}(u_-)]} \right]^2. \quad (34)$$

If $\mathcal{C}(u_+) \neq \mathcal{C}(u_-)$ and the equilibrium radii are positive, then $\Xi_{ij}^{+-} = 1$ if and only if $\beta_i = \beta_j$. Consequently, the universal-response subspace of this completion is the codimension- $(N-1)$ condition $\beta_1 = \dots = \beta_N$.

At fixed nonzero phase charges, Routh reduction adds $\sum_i \pi_i^2 / (2m_i r_i^2)$ to the fast effective potential. The equilibrium radii then acquire smooth charge-dependent corrections whenever the corrected fast Hessian remains nonsingular. Equations (27)–(29) are the leading adiabatic, weak-clock-charge result; they do not assert preparation-independent radial back-reaction.

4.4 Metric, potential, and reduction

Let $m, K, B_V > 0$ and $\mu \geq 0$. Choose

$$V(u) = B_V u^2 (1 - u)^2. \quad (35)$$

The best-matched kinetic metric is

$$d\Sigma^2 = m \sum_{a=1}^3 (dy_a - dg)^2 + \mu dM^2 + K du^2 + I_1(u) d\theta_1^2 + I_2(u) d\theta_2^2. \quad (36)$$

The I_i terms encode persistent organization in the clock sector. The μdM^2 term encodes the additional metric cost of changing active particle-separation records. Exact particle co-motion removes the latter term but not the former.

Before reduction, the constraints are

$$\mathcal{J} = p_1 + p_2 + p_3 \approx 0, \quad (37)$$

$$\mathcal{H} = \frac{1}{2} p^T G_y(u)^{-1} p + \frac{p_u^2}{2K} + \frac{\pi_1^2}{2I_1(u)} + \frac{\pi_2^2}{2I_2(u)} + V(u) - E \approx 0, \quad (38)$$

where $G_y(u) = m\mathbf{1} + \mu L(u)$. Translation invariance gives $\{\mathcal{H}, \mathcal{J}\} = 0$; the two constraints are first class. This is a finite global consistency check, not an analogue of the continuum hypersurface-deformation algebra.

Introduce Jacobi coordinates

$$R = \frac{y_1 + y_2 + y_3}{\sqrt{3}}, \quad \rho = \frac{y_2 - y_1}{\sqrt{2}}, \quad \xi = \frac{y_1 + y_2 - 2y_3}{\sqrt{6}}. \quad (39)$$

The translation constraint is $p_R = 0$. The fixed orthogonal rotation

$$x = \frac{\sqrt{3}\rho + \xi}{2}, \quad z = \frac{-\rho + \sqrt{3}\xi}{2} \quad (40)$$

then gives

$$dM^2 = 3dx^2 + (1 + 2u^2)dz^2. \quad (41)$$

Thus

$$A_x = m + 3\mu, \quad A_z(u) = m + \mu(1 + 2u^2), \quad (42)$$

and the reduced Hamiltonian constraint is

$$\mathcal{H}_{\text{red}} = \frac{p_x^2}{2A_x} + \frac{p_z^2}{2A_z(u)} + \frac{p_u^2}{2K} + \frac{\pi_1^2}{2I_1(u)} + \frac{\pi_2^2}{2I_2(u)} + V(u) - E \approx 0. \quad (43)$$

The cyclic momenta p_x, p_z, π_1, π_2 are constants of motion. Define

$$E_x = \frac{p_x^2}{2A_x}, \quad (44)$$

$$V_{\text{eff}}(u) = V(u) + \frac{p_z^2}{2A_z(u)} + \frac{\pi_1^2}{2I_1(u)} + \frac{\pi_2^2}{2I_2(u)}, \quad (45)$$

$$E_u = E - E_x. \quad (46)$$

A monotonic orbit connects u_- to u_+ whenever

$$E_u > \max_{u \in [u_-, u_+]} V_{\text{eff}}(u). \quad (47)$$

With $\varepsilon = \text{sgn } p_u$, the orbit is determined by quadratures:

$$\sigma - \sigma_0 = \varepsilon \sqrt{\frac{K}{2}} \int_{u_0}^u \frac{dv}{\sqrt{E_u - V_{\text{eff}}(v)}}, \quad (48)$$

$$\theta_i(u) - \theta_{i0} = \varepsilon \pi_i \sqrt{\frac{K}{2}} \int_{u_0}^u \frac{dv}{I_i(v) \sqrt{E_u - V_{\text{eff}}(v)}}, \quad (49)$$

$$z(u) - z_0 = \varepsilon p_z \sqrt{\frac{K}{2}} \int_{u_0}^u \frac{dv}{A_z(v) \sqrt{E_u - V_{\text{eff}}(v)}}. \quad (50)$$

5 Co-motion loading and stationary-action monotonicity

The active-link term is not only a definition of relational change; it contributes to the invariant Jacobi length.

Proposition 3 (Stationary-action monotonicity). *Let γ_μ be a differentiable family of stationary histories with fixed relational endpoints, all within one winding or homotopy class. Assume a regular Jacobi denominator and no branch crossing or caustic on the family. If*

$$G_{AB}^{(\mu)} = G_{AB}^{(0)} + \mu M_{AB}, \quad M_{AB} \dot{q}^A \dot{q}^B \geq 0, \quad (51)$$

then on that stationary branch

$$\frac{\partial S_{\text{on}}}{\partial \mu} = S_0 \int d\lambda \frac{\Omega(q) M_{AB} \dot{q}^A \dot{q}^B}{\sqrt{2\Omega(q) G_{CD}^{(\mu)} \dot{q}^C \dot{q}^D}} \geq 0. \quad (52)$$

For the connected three-particle model, equality holds if and only if $\dot{x} = \dot{z} = 0$ almost everywhere, namely particle co-motion modulo translation.

Proof. Differentiate the action with respect to μ . The implicit variation of the stationary history contributes an Euler-Lagrange term and a fixed-endpoint term, both zero. The remaining explicit derivative gives the displayed expression. Here

$$M_{AB} \dot{q}^A \dot{q}^B = 3\dot{x}^2 + (1 + 2u^2)\dot{z}^2, \quad (53)$$

which vanishes precisely when both reduced shape differentials vanish. $\square$

This is a branchwise envelope-theorem statement. It does not compare independently prepared moving and co-moving experiments, and it does not imply monotonicity of the globally minimizing branch if different stationary branches exchange dominance. Persistent organization $C(u)$ and the link coordinate u may remain nontrivial when the active-link contribution vanishes.

For $p_z \neq 0$, a local clock-versus-shape comparison is

$$R_{iz}(u) = \frac{d\theta_i}{dz} = \frac{\pi_i/I_i(u)}{p_z/A_z(u)} = \frac{\pi_i A_z(u)}{p_z I_i(u)}. \quad (54)$$

The two interaction effects are distinct: $I_i(u)$ carries persistent clock response, while $A_z(u)$ carries active-link particle-non-co-motion loading. Under exact particle co-motion, $p_x = p_z = 0$ and no particle-shape clock is available; the μ term drops out of the remaining clock-link dynamics.

6 Complete observables and relational clock transport

For the transport statements in this section, the displayed two-clock Hamiltonian is extended in the evident way by replacing its clock contribution with $\sum_{i=1}^N \pi_i^2/[2I_i(u)]$. The explicit network orbit and preparation table continue to use $N = 2$.

6.1 Complete observables on monotonic branches

Let $\alpha_{\mathcal{H}}^s$ denote the Hamiltonian flow generated by $\mathcal{H}_{\text{red}}$. On a branch where the lifted phase $T_i = \theta_i \in \widetilde{\mathbb{R}}$ is monotonic,

$$\{T_i, \mathcal{H}_{\text{red}}\} = \frac{\pi_i}{I_i(u)} \neq 0. \quad (55)$$

Let $s_i(x, \tau)$ be the local solution of

$$\alpha_{\mathcal{H}}^{s_i(x, \tau)} T_i(x) = \tau. \quad (56)$$

For a phase-space function f , define

$$F_{f|T_i}(\tau)(x) = \alpha_{\mathcal{H}}^{s_i(x, \tau)} f(x). \quad (57)$$

Lemma 4 (Dirac property). *On every branch on which T_i is monotonic and the inverse is single valued,*

$$\{F_{f|T_i}(\tau), \mathcal{H}_{\text{red}}\} \approx 0. \quad (58)$$

Proof. A gauge displacement by ϵ changes the required flow parameter by

$$s_i(\alpha_{\mathcal{H}}^\epsilon x, \tau) = s_i(x, \tau) - \epsilon. \quad (59)$$

The complete observable is therefore constant along the Hamiltonian gauge orbit. Differentiation at $\epsilon = 0$ gives the result. $\square$

Using the quadrature above, define $\Theta_i(u; u_0)$ as the lifted phase along the orbit. Then

$$F_{u|T_i}(\tau) = \Theta_i^{-1}(\tau; u_0), \quad F_{T_j|T_i}(\tau) = \Theta_j(\Theta_i^{-1}(\tau; u_0); u_0). \quad (60)$$

On a segment where both clocks are monotonic, selecting an event with T_1 and then changing to T_2 selects the same point on the same gauge orbit. This is the standard complete-observable consistency condition, not a new theorem of clock neutrality.

The same construction applies when u is monotonic. Although u is invariant under translations, it changes along Hamiltonian gauge flow and is therefore a partial observable. For any phase-space scalar r , define

$$F_{r|u}(u_*) = \alpha_{\mathcal{H}}^{s_u(x, u_*)} r(x), \quad \alpha_{\mathcal{H}}^{s_u(x, u_*)} u(x) = u_*. \quad (61)$$

It obeys $\{F_{r|u}(u_*), \mathcal{H}_{\text{red}}\} \approx 0$ on the chosen monotonic branch.

6.2 Kinematic identity versus the observability result

The phase-shift symmetries generate conserved Noether charges

$$Q_i = S_0 \pi_i = S_0 I_i(u) \frac{\dot{\theta}_i}{N}, \quad \frac{dQ_i}{d\lambda} = 0. \quad (62)$$

Let

$$\eta_i = \frac{d\theta_i}{2\pi}, \quad \oint_{S_i^1} \eta_i = 1 \quad (63)$$

be the normalized cycle one-form. Its pullback to an orbit gives the local cycle rate $\nu_i = \dot{\theta}_i/(2\pi)$. For $i \neq j$, the local ratio

$$r_{ij} = \frac{\nu_i}{\nu_j} = \frac{\dot{\theta}_i}{\dot{\theta}_j} = \frac{\pi_i I_j(u)}{\pi_j I_i(u)} \quad (64)$$

is a partial observable. Equation (64) is the familiar variable-inertia identity. By itself it neither solves the cross-sector calibration problem nor proves that a clock comparison is gauge invariant. The transport result begins only after clock identity, connected transport, relational event selection, and preparation are supplied.

On a branch where u is monotonic, define the conditional complete observable

$$R_{ij|u}(u_*) := F_{r_{ij}|u}(u_*) = \frac{\pi_i I_j(u_*)}{\pi_j I_i(u_*)}, \quad \{R_{ij|u}(u_*), \mathcal{H}_{\text{red}}\} \approx 0. \quad (65)$$

Theorem 5 (Connected relational clock transport). *Consider one regular gauge orbit that crosses the relational surfaces $u = u_-$ and $u = u_+$. Suppose that (i) the N compact clock actions, orientations, and normalized unit cycles are fixed throughout the connected history; (ii) u is monotonic between the selected surfaces; and (iii) no intervention changes the nonzero conserved phase charges. Then, for every ordered pair $i \neq j$,*

$$\Xi_{ij}^{+-} := \frac{R_{ij|u}(u_+)}{R_{ij|u}(u_-)} = \frac{I_j(u_+)/I_j(u_-)}{I_i(u_+)/I_i(u_-)} \quad (66)$$

is a branch-local Dirac observable, independent of the arbitrary orbit parameter and of the initial charge ratio π_i/π_j .

$$\ln \Xi_{ij}^{+-} = d_j - d_i, \quad d_i := \ln I_i(u_+) - \ln I_i(u_-). \quad (67)$$

Thus the ij transport is nontrivial exactly when clocks i and j undergo unequal fractional changes.

Proof. Each $R_{ij|u}(u_*)$ is a complete observable and hence is constant on the Hamiltonian gauge orbit. Its endpoint ratio is therefore also a Dirac observable. Conservation of the same π_i and π_j between the two relational events cancels their charge ratio and gives (66). Taking the logarithm gives (67). $\square$

Proposition 6 (Calibration quotient). *For N transported clocks, the pairwise observables satisfy*

$$\Xi_{ji}^{+-} = (\Xi_{ij}^{+-})^{-1}, \quad \Xi_{ij}^{+-} \Xi_{jk}^{+-} = \Xi_{ik}^{+-}. \quad (68)$$

There are therefore at most $N - 1$ independent logarithmic transport observables. Their complete endpoint data are represented by

$$[d] \in \mathbb{R}^N / \text{span}\{\mathbf{1}\}, \quad \mathbf{1} = (1, \dots, 1). \quad (69)$$

Every pairwise transport factor is unity if and only if $d \in \text{span}\{\mathbf{1}\}$.

Proof. The relations (68) follow immediately from (67). Choosing one reference clock gives the $N - 1$ coordinates $d_i - d_1$, $i = 2, \dots, N$. Adding the same number to every d_i leaves all such differences unchanged, and these are the only vectors in the kernel of the difference map. Hence the observable data are exactly the quotient (69). $\square$

The theorem and quotient proposition are the principal transport results. The elementary identity (64) is necessary input, but the content is that the endpoint comparison remains

meaningful after gauge, calibration, cycle-identification, and protocol requirements are imposed. For the two-clock model below we write $\Xi_{+-} := \Xi_{12}^{+-}$.

For the linear effective ansatz,

$$\Xi_{+-} = \frac{[M_2 + g_2 C_+][M_1 + g_1 C_-]}{[M_1 + g_1 C_+][M_2 + g_2 C_-]}, \quad C_{\pm} = C(u_{\pm}), \quad (70)$$

and, when $C_+ \neq C_-$,

$$\Xi_{+-} \neq 1 \iff g_2 M_1 \neq g_1 M_2. \quad (71)$$

In the weak-coupling regime,

$$\ln \Xi_{+-} = \left(\frac{g_2}{M_2} - \frac{g_1}{M_1} \right) (C_+ - C_-) + O(g_i^2). \quad (72)$$

The surface $g_1/M_1 = g_2/M_2$ is the exact relative-clock no-effect condition for this ansatz. Organization may still affect the common geometry and link dynamics even though the differential two-clock signal vanishes. In the present diagonal model the transport is integrable and endpoint dependent. No nontrivial clock-bundle holonomy is claimed; path dependence would require additional nonintegrable or off-diagonal clock structure.

6.3 Preparation dependence as a principal result

The same connected model yields different endpoint comparisons under different interventions. Table 2 makes the physical input explicit.

Protocol	Canonical preparation	Endpoint double ratio
Uninterrupted transport	Same clocks; conserved π_1/π_2 on one orbit	Ξ_{+-}
Rate resetting	Same imposed angular-rate ratio at each endpoint	1
Equal clock energies	Independently prepared $E_i^{\text{cl}} = \pi_i^2/(2I_i)$	$\sqrt{\Xi_{+-}}$

Table 2: Preparation protocols and endpoint double ratios.

Preparation dependence does not make Ξ_{+-} gauge dependent. It means that distinct experimental interventions define distinct relational comparisons. The finite claim is specifically that uninterrupted Noether-charge transport through one connected organization change produces Eq. (66).

7 Operational invariance and geometric nonredundancy

7.1 Fixed-organization slices

At a fixed $u = u_*$, the reduced kinetic metric has constant coefficients,

$$A_x dx^2 + A_z(u_*) dz^2 + I_1(u_*) d\theta_1^2 + I_2(u_*) d\theta_2^2. \quad (73)$$

The noncompact shape block can be flattened by constant linear rescalings, and the clock block is a flat rectangular torus. A fixed slice by itself therefore cannot establish organization-dependent transport. The comparison must involve one connected history across different u surfaces.

7.2 Clock-preserving equivalence and non-removability

The operational unit cycle is preserved under phase-origin shifts and $U(1)$ -equivariant isometries that preserve the normalized generator and cycle orientation. A formal local rescaling

$$\tilde{\theta}_i = A_i(u)\theta_i, \quad A_i(u) = \sqrt{\frac{I_i(u)}{I_i(u_0)}} \quad (74)$$

produces

$$d\tilde{\theta}_i = A_i d\theta_i + A'_i \theta_i du. \quad (75)$$

It does not flatten the connected metric without introducing u -clock cross terms and explicit angular-chart dependence.

The global circle issue can be stated exactly. For a compact phase with fixed 2π normalization, the linear map $[\theta] \mapsto [A\theta]$ descends to the quotient $\mathbb{R}/(2\pi\mathbb{Z})$ only when $A \in \mathbb{Z}$. When $|A| > 1$ it is an $|A|$ -fold covering map and is not a diffeomorphism; $A = -1$ reverses orientation. Thus the only orientation-preserving degree-one linear rescaling is $A = 1$.

Proposition 7 (Non-removability under clock-preserving equivalence). *Fix a normalized $U(1)$ clock action with generator $k_i = \partial_{\theta_i}$, orientation, and unit cycle $\theta_i \sim \theta_i + 2\pi$. If the fiber circumference*

$$L_i(u) = \oint_{S_i^1} \sqrt{G(k_i, k_i)} d\theta_i = 2\pi \sqrt{I_i(u)} \quad (76)$$

is nonconstant, then no degree-one orientation-preserving diffeomorphism that preserves the specified clock action and unit cycle can transform the connected clock metric to one with constant fiber circumference.

Proof. A clock-preserving equivalence maps the normalized generator and its degree-one closed orbit to the corresponding normalized generator and degree-one orbit. The Riemannian length of that specified orbit is invariant under pullback by an isometry. Hence a family with nonconstant $L_i(u)$ cannot be isometric, under the allowed equivalence, to one with constant circumference. The only linear orientation-preserving degree-one rescaling of the normalized circle is $A = 1$; noninteger A is not well defined on the quotient, and integer $|A| > 1$ changes the winding degree. A general u -dependent angular coordinate change cannot evade the length invariant and generically produces mixed $du d\theta_i$ terms. $\square$

The transport ratio can be written

$$\Xi_{+-} = \frac{L_2^2(u_+)L_1^2(u_-)}{L_1^2(u_+)L_2^2(u_-)}. \quad (77)$$

Thus Ξ_{+-} is invariant under orbit reparameterizations, global particle translations, phase-origin shifts, and isometries preserving the two specified clock-circle actions and their unit cycles. A transformation that changes a fiber circumference or which closed orbit counts as one cycle defines a different clock construction.

7.3 Curvature of the connected metric

Operational cycle structure establishes the clock meaning; curvature provides a separate geometrical obstruction to flattening the connected metric. The organization-sensitive shape plane has

$$ds_{uz, \text{kin}}^2 = K du^2 + A_z(u) dz^2, \quad (78)$$

with sectional curvature

$$K_{uz}^{\text{kin}} = -\frac{A_z''}{2KA_z} + \frac{(A_z')^2}{4KA_z^2} = -\frac{2\mu(m+\mu)}{KA_z(u)^2}. \quad (79)$$

It is strictly negative for $\mu > 0$. For $I_i(u) = a_i + b_i u^2$, the analogous kinetic curvature is

$$K_{u\theta_i}^{\text{kin}} = -\frac{a_i b_i}{KI_i(u)^2}. \quad (80)$$

The actual unparametrized trajectories are geodesics of the Jacobi metric $\tilde{G}_{AB} = F(u)G_{AB}$, with

$$F(u) = 2[E - V(u)]. \quad (81)$$

Its induced (u, z) curvature is

$$K_{uz}^J = -\frac{1}{2F\sqrt{KA_z}} \frac{d}{du} \left[\frac{(FA_z)'}{\sqrt{FKA_z}} \right]. \quad (82)$$

The conformal factor must therefore be included before making a claim about the full Jacobi geometry. Curvature does not by itself identify a variable as a clock; the specified compact cycle does that operational work.

8 Representative connected orbit

Consider the dimensionless parameter set

$$m = K = r_{10} = r_{20} = 1, \quad \mu = 0.4, \quad B_V = 0.2, \quad \kappa = 0.5, \quad (83)$$

$$M_1 = 1.2, \quad M_2 = 1.0, \quad g_1 = 0.12, \quad g_2 = 0.25, \quad E = 5. \quad (84)$$

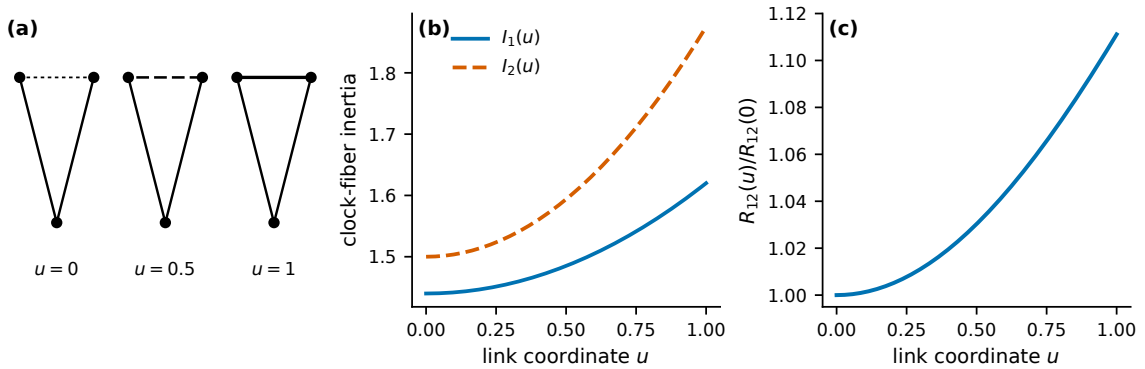


Figure 1: Connected interaction change and relative clock transport. (a) The third interaction channel is continuously activated by the nonnegative weight u^2 while the permanent links keep the graph connected. (b) The illustrative clock-fiber inertias change unequally. (c) The normalized relative cycle-rate ratio changes accordingly and reaches $\Xi_{+-} = 1.111111$ at $u = 1$.

Set $p_x = 0$ and impose the lifted endpoint differences

$$\Delta z = 0.5, \quad \Delta\theta_1 = 0.8, \quad \Delta\theta_2 = 0.55, \quad u_- = 0, \quad u_+ = 1. \quad (85)$$

Solving the endpoint quadratures gives

$$(p_z, \pi_1, \pi_2) = (1.5177135, 2.2376786, 1.6581369). \quad (86)$$

The effective potential remains below $E_u = 5$ throughout the interval, so the branch crosses from $u = 0$ to $u = 1$ without a turning point. The conservative orbit does not settle at $u = 1$; it continues until a later turning point unless dissipation is added.

Figure 1 displays the conceptual content of the example. The closure edge is activated continuously; the two clock inertias change by unequal fractions; and the same-orbit relative cycle ratio therefore changes. The endpoint inertias are

$$(I_1, I_2)|_{u=0} = (1.44, 1.50), \quad (I_1, I_2)|_{u=1} = (1.62, 1.875), \quad (87)$$

and $\Xi_{+-} = 1.111111$. These values are illustrative and are not compared with experiment.

A fixed-endpoint numerical check of the stationary-action proposition resolves the conserved charges after changing μ so that the endpoint data and winding branch remain fixed. At $\mu = 0.4$, a centered finite difference with step 10^{-4} gives

$$\left. \frac{1}{S_0} \frac{\partial S_{\text{on}}}{\partial \mu} \right|_{\text{finite difference}} = 0.35388517, \quad (88)$$

while direct evaluation of the sensitivity integral gives

$$\left. \frac{1}{S_0} \frac{\partial S_{\text{on}}}{\partial \mu} \right|_{\text{sensitivity integral}} = 0.35388518. \quad (89)$$

The agreement verifies the envelope-theorem calculation on this regular branch; it is not evidence for a physical value of μ .

9 Interpretation, limitations, and correspondence

9.1 What the finite model establishes

Within its stated domain, the model establishes five points. First, open and closed interaction organizations can be represented in one connected constrained phase space, avoiding an arbitrary identification between separately calibrated models. Second, uninterrupted transport of N specified compact clocks produces conditional complete observables whose endpoint data form the quotient (69). Third, nonconstant fiber circumference is invariant under the specified class of clock-preserving equivalences. Fourth, persistent organization and particle non-co-motion can occupy distinct metric blocks, and the stationary Jacobi action is branch-wise nondecreasing in the positive co-motion-loading coefficient. Fifth, the fast-mode system (24)–(25) produces a nontrivial clock-fiber metric by pullback from constant bare masses.

The exact Noether transport law is

$$\frac{d}{d\lambda} [I_i(u) \omega_i] = 0, \quad \omega_i = \frac{\dot{\theta}_i}{N}, \quad (90)$$

or equivalently

$$d\omega_i + \omega_i d \ln I_i = 0 \quad (91)$$

along the orbit. This is clock-specific weighted Noether transport. It is not a new connection with nonzero curvature and does not by itself imply a universal weighted-conservation law.

9.2 Universal fractional response and operational invisibility

The universal-response surface marks the limit of what local pairwise clock comparisons can reveal. If every clock has

$$I_i(u) = I_{i0}F(u), \quad (92)$$

then, for every $i \neq j$,

$$R_{ij|u}(u_*) = \frac{\pi_i I_{j0}}{\pi_j I_{i0}}, \quad \Xi_{ij}^{+-} = 1. \quad (93)$$

Each individual rate changes under uninterrupted charge transport, $\omega_i \propto F^{-1}$, but every relative clock ratio is unchanged. A universal multiplicative response is therefore invisible to co-located pairwise clock comparisons within the closed system.

Equivalently, a common fractional response gives $d \in \text{span}\{\mathbf{1}\}$ and hence $[d] = 0$ in (69). The cancellation is therefore a property of the entire pairwise comparison algebra, not a peculiarity of choosing two clocks.

This cancellation is not evidence that F is meaningless in every enlarged theory. It states the additional structure required to give it empirical content. A universal factor could become observable only through a comparison that is not transformed in the same way: for example, transport between spatially separated regions, coupling to an independently specified geometric interval or propagation sector, boundary data, or a non-clock observable with a different response. None of that additional structure is present in the finite model. Consequently, the model does not license interpreting $F(u)$ by itself as an observable scalar “rate of time.”

Three response classes should therefore be kept distinct:

1. *Universal fractional response:* $I_i = I_{i0}F$, for which all pairwise transport signals cancel.
2. *Common factor with representation weights:* $I_i = I_{i0}\Pi(u)^{w_i}$, for which

$$\ln \Xi_{ij}^{+-} = (w_j - w_i)\Delta \ln \Pi. \quad (94)$$

Only differences of weights are observable in relative clock transport.

3. *Clock-specific response:* independent positive $I_i(u)$, for which the general criterion in Theorem 5 applies without a single common factor.

The linear ansatz used here admits the common-factor form only approximately at weak coupling. A more microscopic theory would have to determine whether the universal subspace is an attractor, a symmetry-protected relation, or a tuning.

9.3 What remains model dependent

The induced-metric model proves existence of a stable finite mechanism, not uniqueness or phenomenological realism. Its organization function $\mathcal{C}(u)$ and couplings are prescribed, the phase-only reduction assumes fast radial and mediator modes, and nonzero phase charges perturb the equilibrium radii. The link coordinate and graph weights may ultimately be ordinary collective coordinates and kinetic matrices of a many-body model. Unequal clock responses are therefore not predictions for atomic, optical, nuclear, or mechanical clocks. A more microscopic completion could identify the construction with conventional entrainment or variable-inertia dynamics; the transport and quotient results would remain valid for that system.

Regularity also remains branch dependent. The model requires positive metric coefficients, $E_u > V_{\text{eff}}$, and nonzero conjugate charge for whichever variable is used as a clock. Turning points terminate u as a monotonic label; $p_z = 0$ removes the shape clock; and compact phases cease to be global labels after one cycle unless periodic-clock data are supplied.

9.4 Limits of the finite construction

A continuum extension would require quasi-local organization variables, cluster decomposition, local first-class constraints, a covariant replacement of the graph form, and a common characteristic structure for matter. None of these follows from the finite translation and Hamiltonian constraints used here. In particular, a conserved phase charge is not a local current density, stress-energy tensor, or gravitational source.

The next mathematical step is to derive the organization coordinate and link weights from a reciprocal many-body interaction, rather than prescribe them. The full slow-fast stability matrix and charge-dependent equilibrium manifold could then determine whether the universal-response subspace is dynamically selected or requires parameter tuning.

10 Conclusion

A comparison between separately calibrated interaction sectors cannot by itself establish an intrinsic clock effect. Each sector may admit its own calibration, and the result depends on how endpoint canonical data are identified. The present construction places both organizations, the particle relations, and N operationally specified compact clocks in one connected reparameterization-invariant system. The clocks are transported without intervention, and their pairwise rates at selected organization surfaces are represented by branch-local complete observables.

The content is not the familiar identity $I\dot{\theta} = \text{const}$. After clock identity and unit-cycle normalization are fixed, the same conserved canonical data are transported, and relational organization surfaces and preparation are specified, the cross-organization ratios survive as Dirac observables. Their logarithms are differences $d_j - d_i$; hence only $N - 1$ are independent and their data belong to $\mathbb{R}^N / \text{span}\{\mathbf{1}\}$. A nonconstant fiber circumference also cannot be eliminated by a degree-one clock-preserving diffeomorphism; a covering map or changed cycle normalization defines a different clock construction.

The comparison is not preparation independent. Uninterrupted Noether-charge transport, rate resetting, and equal-energy preparation define different physical experiments. Nor does the model make a universal factor locally visible: if all clock fibers share the same fractional response, every two-clock double ratio cancels. Such a factor could acquire operational meaning only through additional spatial, geometric, propagation, boundary, or non-clock structure not supplied here.

Persistent organization changes the clock-fiber inertias, while departure from particle co-motion enters through a positive active-link graph form. Increasing the coefficient of that form cannot decrease the stationary Jacobi action on a fixed regular branch, with equality precisely for particle co-motion modulo translation. The fast-mode construction additionally shows that variable clock fibers and the corresponding correction to the organization metric can be induced from a stable constant-mass kinetic system. These are finite mathematical-physics results; no universal clock response, local spacetime dynamics, or gravitational theory has been derived.

Appendix A Phase reduction and equations of motion

For $\psi_i = r_i e^{i\theta_i}$,

$$d\psi_i^* d\psi_i = (dr_i)^2 + r_i^2 (d\theta_i)^2. \quad (\text{A1})$$

The metric coefficient in the effective phase-clock ansatz reduces to $I_i(u)$ when the steep radial potential constrains r_i near r_{i0} . Corrections are controlled by the ratio of the organization-

change scale to the radial mass. The reduced einbein equations are

$$\dot{x} = N \frac{p_x}{A_x}, \quad \dot{p}_x = 0, \quad \dot{z} = N \frac{p_z}{A_z(u)}, \quad \dot{p}_z = 0, \quad (\text{A2})$$

$$\dot{u} = N \frac{p_u}{K}, \quad \dot{p}_u = -N \frac{dV_{\text{eff}}}{du}, \quad \dot{\theta}_i = N \frac{\pi_i}{I_i(u)}, \quad \dot{\pi}_i = 0. \quad (\text{A3})$$

Combining the u equation with the constraint yields the orbit quadratures.

Appendix B Diagonalization and curvature

In the (ρ, ξ) basis, the reduced active-link matrix is

$$M(u) = \frac{1}{2} \begin{pmatrix} 5 + u^2 & \sqrt{3}(1 - u^2) \\ \sqrt{3}(1 - u^2) & 3(1 + u^2) \end{pmatrix}. \quad (\text{B1})$$

Its u -independent orthonormal eigenvectors are the combinations defining (x, z) , with eigenvalues 3 and $1 + 2u^2$. Hence

$$\det[m\mathbf{1} + \mu M(u)] = (m + 3\mu)[m + \mu(1 + 2u^2)] > 0. \quad (\text{B2})$$

For a two-dimensional metric $ds^2 = Kdu^2 + A(u)dz^2$, set $r = \sqrt{K}u$ and $f(r) = \sqrt{A(u(r))}$. The Gaussian curvature is $-f''/f$, yielding the curvature quoted in the text.

Appendix C Numerical boundary-value construction

Using u as the monotonic path parameter, collect the cyclic charges as

$$c = (p_z, \pi_1, \pi_2), \quad A(u) = (A_z, I_1, I_2). \quad (\text{C1})$$

Define

$$q_c(u) = \sum_{a=1}^3 \frac{c_a^2}{2[E - V(u)]A_a(u)}. \quad (\text{C2})$$

The stationary endpoint derivatives are

$$\frac{dQ_a}{du} = \frac{c_a}{A_a(u)} \sqrt{\frac{K}{2[E - V(u)][1 - q_c(u)]}}, \quad (Q_1, Q_2, Q_3) = (z, \theta_1, \theta_2). \quad (\text{C3})$$

The charges in the numerical example solve the three endpoint equations

$$\Delta Q_a = \int_0^1 \frac{dQ_a}{du} du. \quad (\text{C4})$$

The stationary abbreviated action is

$$\frac{S_{\text{on}}}{S_0} = \int_0^1 \sqrt{\frac{2K[E - V(u)]}{1 - q_c(u)}} du. \quad (\text{C5})$$

Figure 2 shows the effective-potential margin and the monotonic link orbit used in the numerical check.

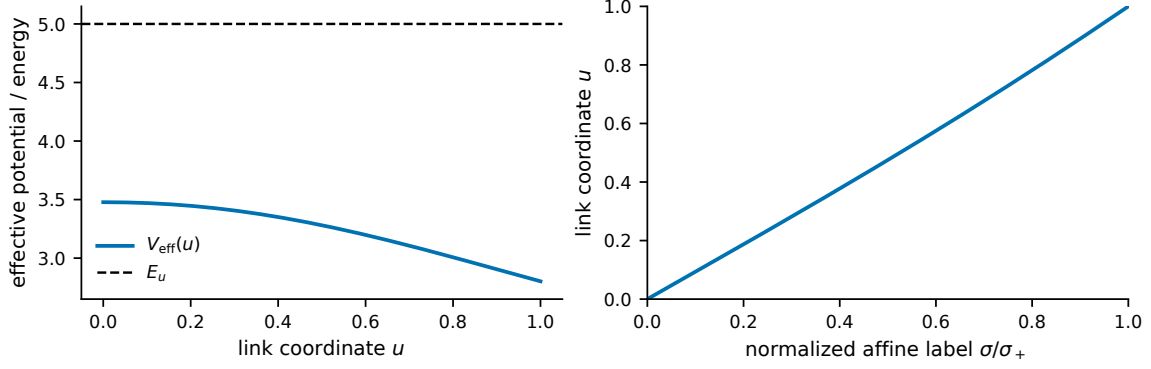


Figure 2: Representative regular branch. The effective potential stays below the available link energy, and the link coordinate is monotonic in the normalized affine label.

Appendix D Equal-energy preparation

At fixed organization, the individual clock energy is

$$E_i^{\text{cl}} = \frac{\pi_i^2}{2I_i}. \quad (\text{D1})$$

If the individual clock energies are prepared equally in independent endpoint experiments, then $\pi_i \propto \sqrt{I_i}$ and

$$\frac{\omega_1}{\omega_2} \propto \sqrt{\frac{I_2}{I_1}}. \quad (\text{D2})$$

The endpoint double ratio is therefore $\sqrt{\Xi_{+-}}$. If angular rates are imposed directly, the double ratio is unity. These alternatives confirm that preparation is physical input rather than gauge.

Supplementary information

Online Resource 1 contains the Python source used to reproduce Figs. 1 and 2, solve the boundary-value problem for the conserved charges, calculate the endpoint transport ratio, and verify the fixed-endpoint sensitivity result. The archived supplementary package is available at <https://doi.org/10.5281/zenodo.22741782>.

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Data availability. No external datasets were used. All numerical inputs are specified in the manuscript.

Code availability. Source code reproducing the figures, conserved charges, endpoint transport ratio, and fixed-endpoint sensitivity check is archived at <https://doi.org/10.5281/zenodo.22741782>.

Use of generative AI. During preparation of this manuscript, the author used OpenAI ChatGPT to assist with structural editing, mathematical exposition, LaTeX preparation, and figure-generation code. The author reviewed and revised all output and takes full responsibility for the content of the manuscript.

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